

DNA/RNA consistency check

Caris TN26-279853 | Population-selected SNP panel | Technical addendum | 5 October 2026

Broad allele compatibility was reproduced across a larger panel selected without using this patient’s DNA allele calls. Local discrepancies remain. These results support continued research use of the paired data; they do not establish sample identity, quantify contamination, validate a treatment target, or change the prior biological findings.

What was tested

The frozen public panel contains 165,782 common SNPs on chromosomes 2, 3, 5, 8, 9 and 17. Selection used population frequencies and preselected exon, regional and genomic-grid intervals. Every position remains in the accounting, including those with no qualifying bases. A callable position requires at least 20 unambiguous A/C/G/T query-name units in each assay.

Coverage stage	DNA	RNA
At least one qualifying A/C/G/T unit	62,262	25,707
At least 20 qualifying A/C/G/T units	35,885	4,072
At least 20 in both assays	4,015 shared positions	4,015 shared positions

Result at positions dominated by one DNA allele

Of 4,015 jointly callable positions, 2,873 have one REF or ALT allele accounting for at least 98% of DNA units. RNA carries that same allele at $\geq 90\%$ at 2,860/2,873 positions (99.55%), and at $\geq 98\%$ at 2,760/2,873 (96.07%). None of these 2,873 baseline DNA-dominant positions switches to $\geq 98\%$ of the opposite REF/ALT allele.

DNA category	Positions	RNA same allele $\geq 90\%$	RNA same allele $\geq 98\%$
REF dominant	2,283	2,272	2,189
ALT dominant	590	588	571
Total	2,873	2,860	2,760

Breadth and selection limits

The dominant positions span all six chromosomes, 554 one-megabase bins and 1,194 uniquely assigned GENCODE genes. Only 72 overlap the earlier 335-marker panel. They are still clustered: 549 adjacent gaps are <1 kb, and surviving sites are strongly enriched near annotated exons. They must not be treated as independent random observations.

A coordinate-first subset spaced ≥ 50 kb apart was frozen independently of allele direction. Of its 17,264 selected sites, 165 were jointly callable and 120 were DNA dominant: all 120 retained the same RNA allele at $\geq 90\%$; 115 did so at $\geq 98\%$. This supports spatial breadth, not a probability of identity.

The remaining 1,142 jointly callable positions comprise 1,109 mixed REF/ALT DNA sites and 33 residual mixtures. RNA becomes $\geq 98\%$ REF or ALT at 68 of the mixed-DNA sites. Such allele-fraction shifts are reported separately and are not counted as identity mismatches.

Discrepancies retained for review

The prespecified strong-discrepancy screen required ≥ 50 A/C/G/T units in each assay, an individual base at $\leq 2\%$ in DNA but $\geq 10\%$ and at least 10 units in RNA. Independent per-base counting across all 4,015 sites confirmed six positions. Counts below are query-name units, not unique molecules. Coordinates are GRCh38, 1-based.

Locus	DNA baseline	RNA baseline	RNA after clean-read filter
COL5A2 chr2:189043211	G58	G327, A116	G183, A0
APOD chr3:195568834	G87	G36, C27	G0, C24
EXOC3 chr5:443259	C153	C74, A22, G1	C11, A21, G1
MSH3 / DHFR chr5:80654917	C912	C9, G104	C7, G53
PABPC1 chr8:100705591	A94	A400, G65	A278, G9
PABPC1 chr8:100705604	G101	G358, C78, A1	G218, C31

The clean-read sensitivity excludes soft-clipped reads and requires at least five query bases between the tested base and each aligned end. It changes both evidence and coverage. COL5A2's unexpected A support disappears; APOD and EXOC3 remain discrepant but fall below depth 50. Falling below a threshold is not resolution. The two PABPC1 positions are only 13 bases apart.

Read-context review

COL5A2's unexpected A support is strongly dependent on read ends. A bounded search of selected representative sequences also found local gap-placement alternatives at APOD, EXOC3 and DHFR: first-best nominated ordinary alignments placed the queried base in an insertion or at a nearby reference position. These alternatives sometimes scored below the original splice-path model containing the unexpected allele. They demonstrate ambiguity, not a proven correction or preferred read origin.

The PABPC1 discrepancies remain unresolved. All 65 baseline query names supporting G at chr8:100705591 also support C at chr8:100705604 in the same qualifying alignment. The two sites therefore share substantial read support and cannot be treated as independent checks. Representative sequences are not independent molecules; the bounded reference search cannot establish allele origin.

Other exceptions and mapping sensitivity

All 13 dominant-DNA sites with $< 90\%$ same-allele RNA retention are preserved in the evidence table, including seven below the strong screen's depth requirement. Raising MAPQ to 60 introduces three further flags—FANCD2, SDHAP2/MUC20-OT1 and FOXD4L4—by removing DNA allele support while leaving RNA unchanged. These are sensitivity-only findings, not replacements for the baseline results.

No discrepancy has been converted into a new tumor mutation, neoantigen, diagnosis or drug recommendation. Locus-specific biology and mapping effects are not separated by the aggregate consistency result.

Methods, verification and scope

Frozen before counting

Design SHA256: 90a153f630375103363792037618d93a864b01bad029575f36d6a3a382ee0bfa

Population selection used 1000 Genomes 2019 GRCh38 biallelic PASS variants (global ALT frequency 0.05–0.95), protein-coding gene exons plus 100 bp, designated gene regions plus 250 kb, and a common-SNP grid. The six-chromosome scope was inherited from the prior copy-number work. SAS frequency was recorded, not used for selection. Every selected REF matched the resident GRCh38 FASTA.

Both original Caris BAMs were recounted with samtools 1.24 / pysam 0.24.1. Filters: MAPQ ≥ 30 ; base quality ≥ 25 ; proper pairs required; flags 0xF0C excluded; BAQ and samtools mate-overlap adjustment disabled. Qualifying observations were collapsed by query name; conflicting A/C/G/T mates were discarded. Each original BAM has one read group. Reference skips, deletions and ambiguous bases do not count toward A/C/G/T depth.

The explicit depth limit was disabled (-d 0, effectively unlimited in samtools). Installed behavior was qualified with 9,001 synthetic paired names / 18,002 records, a finite-cap contrast, and separate base-quality, mapping-quality, duplicate, supplementary and conflicting-mate controls. Pileup columns and BED coordinate conventions were checked before patient counting.

Independent checks

A separate CIGAR-walk implementation reproduced all 4,015 jointly callable DNA and RNA positions: REF/ALT/other query-name counts, conflicts and REF/ALT read counts. Exact integer arithmetic independently reproduced categories, retention thresholds, full-panel attrition and coordinate-first spacing. Public GENCODE annotation passed an independent 81-position boundary/category check.

The fresh uncapped DNA run exactly matched all 935,370 compared integer fields across the 62,358 previously observed DNA rows. Thus the old finite cap had no observed effect on these final counts. This conclusion comes from an actual uncapped recount, not the fact that reported post-quality depth was below the old cap.

Newer RNA alignment: not evaluable here

The newer full-reference STAR run was streamed and did not retain a complete BAM. Its retained BAP1/RASA1/LATS1/LATS2 BAM cannot substitute for the full panel. Only 73 selected public SNPs intersect BAP1/RASA1 retained intervals; none met joint callability (maximum original RNA depth 9). No per-site newer-alignment comparison was performed, and thresholds were not relaxed. The broad result therefore depends on the original Caris alignments.

Interpretation and files

This is a research consistency analysis on selected, coverage-surviving loci—not a validated identity assay, chance-match calculation, contamination estimate, matched-normal analysis, or proof of allele origin. Query names are not UMIs. The original four follow-ups and v2 handoff remain unchanged; this addendum adds technical quality evidence, not a treatment recommendation.

The accompanying evidence archive includes the frozen design, counts for all selected loci, coverage and discrepancy tables, independent audit code/results, read-context evidence, source provenance and hashes. Large compressed raw pileups remain in the resident analysis folder. Local execution only; no new AWS compute resources or external outreach.

Method reference: <https://www.htslib.org/doc/samtools-mpileup.html> (samtools 1.24). GENCODE37 annotation and 1000 Genomes panel provenance are recorded in the evidence files.